One-sided Fourier cones: reality, half-energy and Hermitian interactions

Project Stardust

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Abstract

One-sided Fourier support has different consequences for real fields, complex bilinear tests and Hermitian energy tests. We prove that a real Schwartz field supported in a narrow single cone is zero, and that at most half of each nonnegative radial spectral moment of a real field can lie in such a cone. Complex cone fields can be nonzero and divergence free, although their unconjugated bilinear and advection trilinear integrals vanish. For every cone aperture $0 < \delta \le 1$, we reconstruct explicit smooth compact spectral bumps giving three nonzero divergence-free complex Schwartz fields in that same cone whose Hermitian advection pairing is strictly negative. The proof includes the actual double Fourier integral, cone support under convolution, a positive central-frequency coefficient and a curl response whose pairing is exactly a negative square. These are verified diagnostic constructions and classical Fourier mechanisms, not a new fluid regularity or singularity theorem. Nonvanishing does not disprove an aperture-dependent estimate with a positive constant.

1 Conventions and the diagnostic question

For scalar or vector Schwartz functions on $\mathbb{R}^3$, use

$$\widehat{f}(\xi) = \int_{\mathbb{R}^3} e^{-2\pi i x\cdot\xi} f(x)\,dx, \qquad \kappa = 2\pi i, \qquad \widehat{\partial_j f}(\xi) = \kappa \xi_j \widehat{f}(\xi).$$

Here coordinates are numbered $1, 2, 3$; the formal source numbers them $0, 1, 2$. The norm on $\mathbb{C}^3$ is its Hermitian Euclidean norm. For a unit vector n and $0 < \delta \le 1$, define the open angular cone

$$C(n, \delta) = \{\xi \neq 0 : |\xi/|\xi| - n| < \delta\}. \tag{1}$$

The closed support of a continuous Fourier transform means the closure of its nonzero set. Statements using closed support are made explicitly below.

The issue is which pairing a cancellation actually controls. For scalar fields set

$$B(f, g) = \int fg, \qquad H(f, g) = \int f\overline{g}.$$

For vector fields put $N(u, v) = (u \cdot \nabla)v$ and

$$T_0(u, v, w) = \int \sum_j N(u, v)_j w_j,$$

$$T_H(u, v, w) = \int \sum_j N(u, v)_j \overline{w_j}.$$

These are actual absolutely convergent integrals for Schwartz inputs. In particular, $H(f,f) = \|f\|_2^2$, whereas $B(f,f)$ need not be an energy. The main construction proves $T_0 = 0$ and $T_H < 0$ for the same three fields. It does not make u, v, w identical or construct a time-dependent PDE solution.

The frozen verified baseline is

`bab688bb6140cd6316c80680095a914ea0d346db`.

Exact source declarations, original hypotheses and hashes accompany the paper. All Fourier inversion, Plancherel and Schwartz-space facts used here are classical foundational inputs; the argument reconstructs how the mapped verified sources combine them. No mathematical source was changed.

2 Real fields cannot occupy only one cone

Lemma 2.1 (Antipodes and a positive halfspace)**.** *The cone $C(n,\delta)$ contains no pair $\xi, -\xi$. It is contained in $\{\xi : \xi \cdot n > 0\}$. Moreover, for $\xi \neq 0$,*

$$\xi \in C(n,\delta) \quad \Longleftrightarrow \quad (2-\delta^2)|\xi| < 2\xi \cdot n. \tag{2}$$

The sum of any two elements of the cone belongs to the cone.

Proof. For a unit a, if $|a-n| < \delta$ and $|-a-n| < \delta$, the triangle inequality gives $2 = |2a| < 2\delta \leq 2$, a contradiction. Also $|a-n|^2 = 2 - 2a \cdot n$. Substitution of $a = \xi/|\xi|$ gives (2), and $2-\delta^2 \geq 1$ gives $\xi \cdot n > 0$. If p, q are in the cone, then $(p+q)\cdot n > 0$, so $p+q \neq 0$. Using the triangle inequality and $2-\delta^2 > 0$,

$$(2-\delta^2)|p+q| \leq (2-\delta^2)(|p|+|q|) < 2(p+q)\cdot n.$$

Equation (2) completes the proof. □

Source correspondence: `SingleConeReality`
`cone_no_antipodes; zero_not_mem_cone`

Source correspondence: `ConeSupportAlgebra`
`cone_linear_iff; cone_add_mem`

Theorem 2.2 (Reality and support)**.** *If a scalar real-valued Schwartz function has Fourier nonzero set contained in $C(n,\delta)$, it is zero. The same is true componentwise for a real vector field.*

Proof. Conjugating the defining Fourier integral gives $\widehat{f}(-\xi) = \overline{\widehat{f}(\xi)}$ for real f. Thus if $\widehat{f}(\xi) \neq 0$, then $\widehat{f}(-\xi) \neq 0$ as well. The support hypothesis would put both antipodes in the cone, contrary to Lemma 2.1. Hence $\widehat{f} = 0$, and Fourier inversion gives $f = 0$. Apply this argument to each component for vector fields. □

Source correspondence: `SingleConeReality`
`fourier_reality; real_schwartz_single_cone_zero`

Source correspondence: `SingleConeReality`
`real_vector_schwartz_single_cone_zero`

Thus a nonzero single-cone example in the rest of the paper is necessarily complex. Taking its real part adds reflected Fourier support and changes the single-cone hypothesis.

3 A half-energy bound for real fields

For $c \geq 0$, integer $k \geq 0$ and a vector Schwartz field F, let

$$E_k(F) = \int (c|\xi|)^k |\widehat{F}(\xi)|^2 \, d\xi, \qquad E_{k,C}(F) = \int_C (c|\xi|)^k |\widehat{F}(\xi)|^2 \, d\xi.$$

The density is integrable by Schwartz decay and nonnegative. When $k = 0$, the radial weight is one, including the case $c = 0$.

Theorem 3.1 (At most half in one cap)**.** *For every real-valued vector Schwartz field F,*

$$2E_{k,C}(F) \leq E_k(F).$$

If $E_k(F) > 0$ and $(1-\varepsilon)E_k(F) \leq E_{k,C}(F)$, then $\varepsilon \geq 1/2$. In particular, concentration of more than half of such a radial energy in a single cap is impossible.

Proof. Reality gives $|\widehat{F}(-\xi)|^2 = |\widehat{F}(\xi)|^2$. The radial weight is even, so the energy density is even. Reflection preserves Lebesgue measure; hence the integrals over C and $-C$ are equal. These two measurable sets are disjoint by Lemma 2.1. Their union therefore has energy $2E_{k,C}(F)$, at most the total energy by nonnegativity. Combining this with the assumed lower bound gives $2(1-\varepsilon)E_k(F) \leq E_k(F)$; division by the positive total gives $\varepsilon \geq 1/2$. □

Source correspondence: `SingleConeEnergyLimit cone_twice_le_total; dominance_forces_half_error`

The measurable Fourier projection gives the same physical-space conclusion. Indeed let

$$G_C = \mathcal{F}^{-1}\Big(1_C (c|\xi|)^{k/2} \widehat{F}(\xi)\Big) \quad \text{in } L^2.$$

The multiplier is square integrable by the stated moment bound. Unitarity gives $\|G_C\|_2^2 = E_{k,C}(F)$, so $2\|G_C\|_2^2 \leq E_k(F)$. The projected field is an actual L^2 class; the sharp indicator of the cone is not assumed to preserve Schwartz space.

Source correspondence: `SingleConeEnergyLimit projectedPhysicalField_norm_sq; real_cap_projection_half`

For completeness, the factor $1/2$ can be attained. This elementary sharpness observation is a human deduction, not a named new formal theorem. Choose a nonzero smooth compactly supported Fourier bump b in C, and let $f = \mathcal{F}^{-1}b$. The real function $g = \operatorname{Re} f$ has

$$\widehat{g}(\xi) = \tfrac{1}{2}\big(b(\xi) + \overline{b(-\xi)}\big).$$

The two summands have disjoint supports in C and $-C$. For every nonnegative even radial weight with a positive finite integral against $|b|^2$, each reflected cap contributes exactly one half of the total weighted energy. In particular this applies to the displayed moments for $c > 0$, and also to the unweighted moment. It does not produce a real field supported only in C.

4 What the unconjugated cancellations say

For scalar Schwartz functions, Fourier inversion and Plancherel give

$$B(f,g) = \int \widehat{f}(\xi)\widehat{g}(-\xi) \, d\xi, \qquad H(f,g) = \int \widehat{f}(\xi)\overline{\widehat{g}(\xi)} \, d\xi. \tag{3}$$

The first follows, for example, by applying the second to $f, \overline{g}$ and using $\widehat{\overline{g}}(\xi) = \overline{\widehat{g}(-\xi)}$. Products on both sides are integrable by Cauchy–Schwarz. If f, g have Fourier nonzero sets in C, the first integrand is identically zero because the cone contains no antipodes. Thus $B(f, g) = 0$. For a nonzero f, however, $H(f, f) = \|f\|_2^2 > 0$.

Source correspondence: `SingleConePairings`
`bilinear_fourier_reflection; scalar_hermitian_fourier`

Source correspondence: `SingleConePairings`
`same_cone_bilinear_zero; hermitian_self_positive`

The convolution identity

$$\widehat{fg}(\xi) = \int \widehat{f}(p)\widehat{g}(\xi - p)\, dp$$

shows that the Fourier nonzero set of fg lies in the sum of the two Fourier nonzero sets. If all frequencies of f, g lie in the positive halfspace $\xi \cdot n > 0$, so do the frequencies of fg. Applying (3) to fg and a third function h in that halfspace gives $\int fgh = 0$. Differentiation only multiplies Fourier coefficients by $\kappa\xi_j$, and hence cannot enlarge the Fourier nonzero set. Coordinatewise application therefore proves

$$T_0(u, v, w) = 0 \tag{4}$$

whenever all three vector Schwartz fields have Fourier nonzero sets in C. Solenoidality is not required for this cancellation.

Source correspondence: `SingleConeTrilinear`
`fourier_product_support; positive_triple_zero; cone_trilinear_zero`

In frequency terms the unconjugated triple asks for $p + q + r = 0$, impossible when all three have positive n component. The Hermitian triple asks for $p + q = r$, which can occur inside the same cone. The next section justifies this latter integral as an actual product-space integral before using it.

5 The actual Hermitian Fourier formula

For scalar Schwartz f, g, h,

$$\int fg\overline{h} = \iint \widehat{f}(p)\widehat{g}(q)\overline{\widehat{h}(p+q)}\, dp\, dq. \tag{5}$$

Here the product-space integrand is absolutely integrable, since

$$\iint |\widehat{f}(p)\widehat{g}(q)\widehat{h}(p+q)|\, dp\, dq \le \|\widehat{h}\|_\infty \|\widehat{f}\|_1 \|\widehat{g}\|_1 < \infty.$$

Plancherel applied to fg, h gives $\int \widehat{fg}(\xi)\overline{\widehat{h}(\xi)}\, d\xi$. Insert the convolution formula. The change of variables $(\xi, p) \mapsto (p, q = \xi - p)$ has absolute determinant one. The displayed absolute-integrability bound, transported through that invertible linear map, justifies Fubini and the change of variables, proving (5). No unspecified iterated distributional integral is used.

Source correspondence: `HermitianTriadFourier`
`triadKernel_integrable; frequencyShear_preserving`

Source correspondence: `HermitianTriadFourier`
`scalar_fourier_representation_prod`

Apply (5) to each $u_i, \partial_i v_j, w_j$ and sum the nine terms. This gives the exact vector formula

$$T_H(u,v,w) = \iint \kappa \left(\sum_i \widehat{u}_i(p) q_i \right) \left(\sum_j \widehat{v}_j(q) \overline{\widehat{w}_j(p+q)} \right) dp\, dq. \tag{6}$$

Each summand is absolutely integrable by the previous bound with $\widehat{\partial_i v_j}(q) = \kappa q_i \widehat{v}_j(q)$, still Schwartz. Finite summation thus commutes with integration. This identifies precisely where the conjugate and the sum frequency belong.

Source correspondence: `HermitianTriadFourier vectorKernel_integrable; vector_fourier_representation_prod`

6 A compact spectral bump and two solenoidal inputs

We now construct a nonzero Hermitian interaction for each fixed aperture. Take $n = e_1$ and $0 < \delta \le 1$. Since $C(e_1, \delta)$ is an open neighborhood of e_1, choose a real smooth compactly supported bump b such that

$$0 \le b \le 1, \quad b(e_1) = 1, \quad A := \operatorname{supp} b \Subset C(e_1, \delta). \tag{7}$$

Here $\operatorname{supp} b$ is closed support and the notation means compact containment. A usual smooth bump on a sufficiently small ball about e_1 supplies it. Let $f = \mathcal{F}^{-1} b$, a scalar Schwartz function. Define the two curl-plane operators

$$\mathcal{C}_{12} f = (\partial_2 f, -\partial_1 f, 0), \qquad \mathcal{C}_{13} f = (\partial_3 f, 0, -\partial_1 f),$$

and set

$$u = \mathcal{C}_{12} f, \qquad v = \mathcal{C}_{13}(\partial_2 f). \tag{8}$$

Mixed derivatives commute, so both inputs are divergence free. They are Schwartz because differentiation preserves Schwartz space. Their Fourier transforms are

$$\widehat{u}(p) = \kappa b(p)(p_2, -p_1, 0), \tag{9}$$

$$\widehat{v}(q) = \kappa^2 b(q) q_2 (q_3, 0, -q_1). \tag{10}$$

In particular, both closed Fourier supports lie in A.

Source correspondence: `ConeInteractionCore spectralBump_nonempty; inputU_tsupport; inputV_tsupport`

Source correspondence: `DivergenceFreeConeWitness curl_divergence_zero; fourier_curl`

Source correspondence: `ConeInteractionCore curl02_divergence_zero; fourier_curl02`

Let $N = N(u,v)$. All its components are products of Schwartz functions and derivatives, so N is Schwartz. The convolution formula gives

$$\widehat{N}_j(\xi) = \int \kappa \left(\sum_i \widehat{u}_i(p)(\xi - p)_i \right) \widehat{v}_j(\xi - p)\, dp. \tag{11}$$

Every convolution term is supported in $A + A$. Since A is compact, $A + A$ is compact and closed. Vanishing outside $A + A$ therefore also puts the *closed* Fourier support of N in $A + A$. Lemma 2.1

places that set inside the same cone. Compactness is useful here: merely putting a nonzero set in an open cone would not automatically put its closure there.

Source correspondence: `ConeSupportAlgebra`
`advect_tsupport_subset_sum; advection_tsupport_cone`

7 A strictly positive central coefficient

The inputs were chosen so that one component of (11) has a nonnegative integrand at $\xi_* = 2e_1$. For $q = \xi_* - p$,

$$p_2 q_1 - p_1 q_2 = 2p_2, \qquad q_1 = 2 - p_1, \qquad q_2 = -p_2.$$

Substitute (9)–(10) into the $j = 3$ integrand of (11). Direct multiplication gives

$$\widehat{N}_3(2e_1) = 2\kappa^4 \int p_2^2 (2 - p_1) b(p) b(2e_1 - p)\, dp. \tag{12}$$

There are four derivative factors; in particular $\kappa^4 = (2\pi)^4 > 0$ is real. Losing a factor i here would change the sign test.

Source correspondence: `ConeInteractionCore`
`central_kernel_identity; central_fourier_real_identity`

Lemma 7.1 (Strict positivity)**.** *The integral in* (12) *is finite and strictly positive. Consequently* $\widehat{N}_3(2e_1) > 0$ *is real and* $N \neq 0$*.*

Proof. The integrand is smooth with compact support, so it is integrable. It is nonnegative everywhere. Indeed if $b(2e_1 - p) = 0$ the assertion is immediate; otherwise $2e_1 - p$ is in the cone and hence has positive first coordinate, so $2 - p_1 > 0$. All other factors are nonnegative.

To obtain strict positivity, let $G(p) = (2 - p_1) b(p) b(2e_1 - p)$. It is continuous and $G(e_1) = 1$. For a sufficiently small nonzero t, continuity gives $G(e_1 + te_2) > 0$, while $p_2^2 = t^2 > 0$ at that point. Thus the complete integrand is positive at a point, and by continuity on a nonempty open ball of positive Lebesgue measure. Nonnegativity everywhere and positivity on that ball give a strictly positive integral. Equation (12) and $\kappa^4 > 0$ give the conclusion. □

Source correspondence: `ConeInteractionCore`
`centralDensity_nonneg; centralDensity_integral_pos`

Source correspondence: `ConeInteractionCore`
`central_fourier_positive; advection_nonzero`

This proves nonzero advection without assuming that an arbitrary pair of nonzero cone fields interacts. The particular transverse derivatives in (8) and the positive bump are responsible for the sign.

8 A solenoidal response and an exact negative square

Define the scalar rotor and its response by

$$r = \partial_3 N_1 - \partial_1 N_3, \qquad w = \mathcal{C}_{13} r = (\partial_3 r, 0, -\partial_1 r).$$

They are Schwartz, and w is divergence free by commuting mixed derivatives. Differentiation does not enlarge closed Fourier support, so the closed Fourier supports of r, w lie in $A + A \subset C(e_1, \delta)$. At the central frequency,

$$\widehat{r}(2e_1) = \kappa(2e_1)_3 \widehat{N}_1(2e_1) - \kappa(2e_1)_1 \widehat{N}_3(2e_1) = -2\kappa \widehat{N}_3(2e_1) \neq 0.$$

Thus $r \neq 0$ by Fourier injectivity.

Source correspondence: `ConeInteractionWitness`
`rotor_nonzero; response_physical_divergence; response_tsupport`

Theorem 8.1 (Two different trilinear conclusions)**.** *For every $0 < \delta \leq 1$ there exist three nonzero divergence-free complex Schwartz vector fields u, v, w on $\mathbb{R}^3$, all with closed Fourier support in $C(e_1, \delta)$, such that*

$$T_0(u, v, w) = 0, \qquad T_H(u, v, w) = -\|\partial_3 N(u, v)_1 - \partial_1 N(u, v)_3\|_2^2 < 0.$$

Proof. Use the constructed u, v, w. All support and solenoidality statements were proved above. Integration by parts, justified by Schwartz decay, gives

$$\begin{aligned} T_H(u, v, w) &= \int (N_1 \overline{\partial_3 r} - N_3 \overline{\partial_1 r}) \\ &= -\int (\partial_3 N_1 - \partial_1 N_3)\overline{r} = -\int |r|^2. \end{aligned}$$

The norm is strictly positive: a nonzero continuous Schwartz function is nonzero on an open set of positive measure. Thus $T_H < 0$. If any of u, v, w were zero, its trilinearity would force $T_H = 0$; all three are therefore nonzero. The unconjugated value is zero by (4), since all three Fourier supports lie in the same cone. □

Source correspondence: `ConeInteractionWitness`
`rotor_energy_identity; interaction_real_negative`

Source correspondence: `ConeInteractionWitness`
`exists_full_cone_interaction; pairings_unequal`

In the source Hilbert convention the Hermitian inner product is conjugate linear in its first slot. Its equality is written as pairing w with N; the scalar formula is exactly the T_H displayed here. No switch of inner product convention or silent conjugation is needed.

9 What the example rules out, and what it leaves open

The construction rules out the assertion that positive-frequency cone support alone makes *every* Hermitian advection interaction vanish. It also gives an exact counterexample to identifying the unconjugated physical triple with the conjugated double Fourier integral (6). Those two expressions give zero and a negative number, respectively, on the same inputs.

It does not disprove an inequality with a factor depending on δ. The fields and their norms depend on the chosen bump and aperture; the argument supplies no normalized lower bound uniform as $\delta \downarrow 0$. In particular a genuine small-angle estimate requires its own symbol and norm analysis. An unconjugated zero is a trivial bound for that particular form, and is not by itself an estimate for Hermitian energy.

Nor does this contradict the real divergence-free transport cancellation $\int (u \cdot \nabla) v \cdot v = 0$ under suitable decay. Our inputs are complex and are three independently chosen fields, while nonzero real fields cannot satisfy the single-cone hypothesis of Theorem 2.2. Real-part symmetrization introduces the antipodal cap and changes the frequency interactions. The half-energy theorem quantifies that reality constraint for every displayed radial moment.

The basic conjugation, reflection, Plancherel and convolution mechanisms are classical. Tao's primary Fourier notes [1], including the Schwartz inversion discussion and Theorem 43, provide an accessible foundational comparison. A bounded search for the specific solenoidal compact-bump construction did not establish its priority. No novelty certification follows from that incomplete search; the value of this formalization companion is the complete checked construction and the exact pairing and reality hypotheses. It proves no global regularity, finite-time blowup, nonlinear evolution of cone support or angular partition theorem for the unrestricted Navier–Stokes equations.

References

[1] T. Tao, *245C, Notes 2: The Fourier transform*, April 6, 2009. Primary author exposition, especially Schwartz inversion and Theorem 43 (Plancherel). Author's lecture notes.